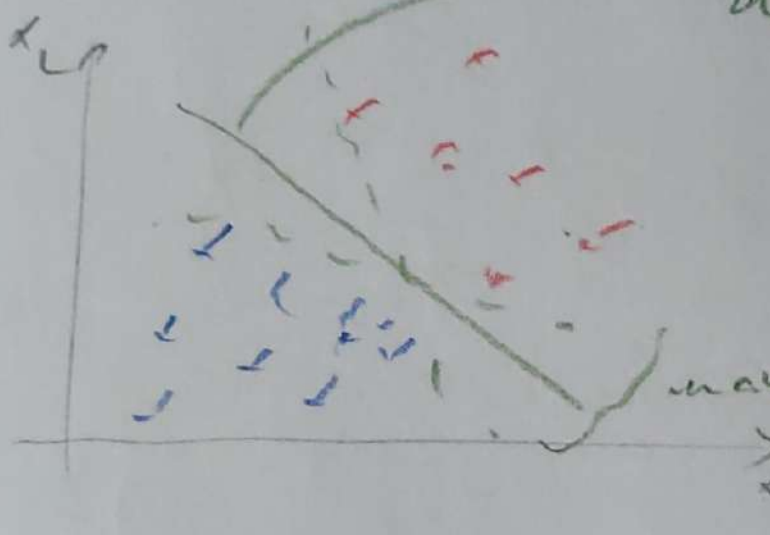
Introduction to machine learning – 5 –

Support vector machines (SVM)

Binary data + logistic regression



decision boundary
hyperplane



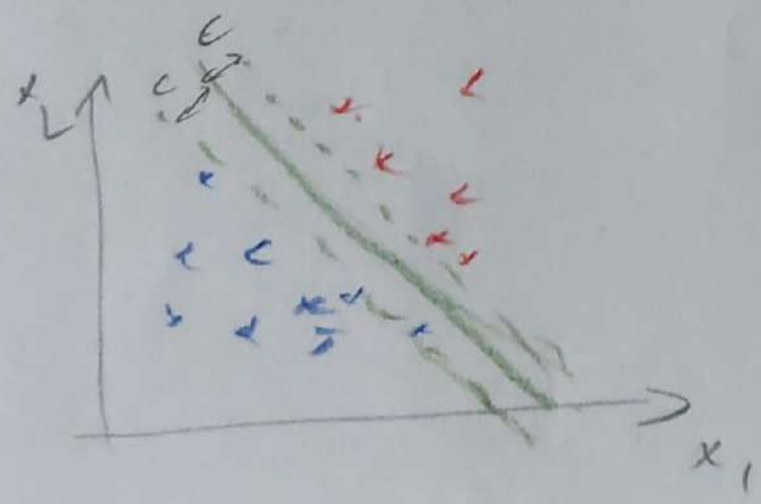
many choices

Which boundary to choose

Equation of the hyperplane:

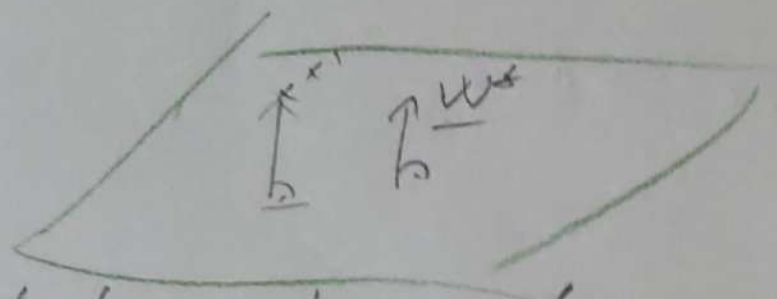
$$f(\underline{x}) = \underline{w}^T \underline{x} + b = 0$$

- Optimal separating hyperplane: maximize margin



hyperplane: $\underline{w}^T \underline{x} + b = 0$ $\underline{w}$ is normal to the plane

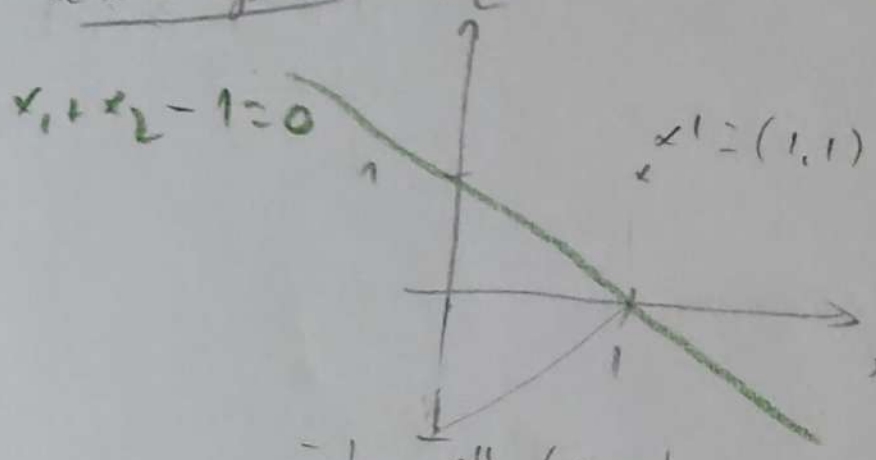
$\underline{w}^* = \frac{\underline{w}}{|\underline{w}|}$ unit vector



distance from plane

$D(x') = \underline{x}' \underline{w}^* + b/|\underline{w}| = \frac{\underline{x}' \underline{w}^T + b}{|\underline{w}|}$ signed distance

example:



$\underline{w} = (1,1)$ $|w| = \sqrt{2}$

$b(x') = \frac{1 \cdot 1 + 1 \cdot 1 - 1}{\sqrt{2}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

$b(x'') = \frac{0 \cdot 1 + -1 \cdot 1 - 1}{\sqrt{2}} = \frac{-2}{\sqrt{2}} = -\sqrt{2}$

Classification:

$\text{sgn}(\underline{w}^T \underline{x}^{(i)} + b) = t^{(i)}$ or $t^{(i)}(\underline{w}^T \underline{x}^i + b) > 0$

If we have a margin (c)

$t^{(i)}(\underline{w}^T \underline{x}^{(i)} + b) \cdot \frac{1}{|w|} \geq c$

Optimization: choose $\underline{w}, b$ to

$\max_{w,b} c$ such that $\frac{1}{|w|} t^{(i)}(\underline{w}^T \underline{x}^{(i)} + b) \geq c$

We have a freedom
$\underline{w}' = 2 \cdot \underline{w}$ is just as good.

Usual choice:
$|\underline{w}| = \frac{1}{c} \Rightarrow t^{(i)}(\underline{w}^T \underline{x}^{(i)} + b) \geq 1$

Optimization:
minimize $|\underline{w}|^2$ for $\underline{w}, b$ such that
$t^{(i)}(\underline{w}^T \underline{x}^{(i)} + b) \geq 1 \quad i = 1 \ldots N$

Methods:
- greedy
- stochastic

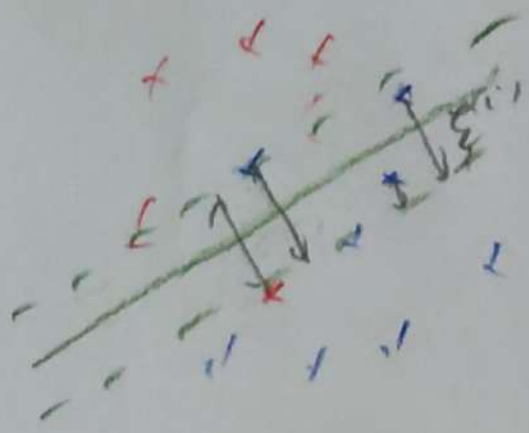
The important training examples that fulfill the equality are called support vectors

Problems: noise, non separable data





Soft margin:
$\min_{\underline{w}, b} \|\underline{w}\|^2 + \lambda \, \text{Error}(\underline{w}, b)$ such that
$t^{(i)}(\underline{w}^T \underline{x}^{(i)} + b) \geq 1$ ← this cannot hold



$\xi^{(i)}$ = error $\quad \xi^{(i)} > 0$

$0 < \xi^{(i)} < 1$: margin violation

$\xi^{(i)} > 1$: misclassification

Minimize $\xi^{(i)}$

Soft-margin SVM

$$\min \frac{1}{2}|w|^2 + \lambda \sum_{i=1}^{N} \xi^{(i)} \quad \text{such that}$$

w.r.t.

$$t^{(i)}(\underline{w}^T \underline{x}^{(i)} + b) \geq 1 - \xi^{(i)}$$

λ hyperparameter

Small λ penalizes errors less, wide margin

large λ narrow margin

$\lambda \to \infty$ hard margin solution

- Eliminate $\xi^{(i)}$

$$\xi^{(i)} \geq 1 - t^{(i)}(\underline{w}^T \underline{x}^{(i)} + b)$$

Two cases:

$1 - t^{(i)}(\underline{w}^T \underline{x}^{(i)} + b) \leq 0$ $\qquad$ $1 - t^{(i)}(\underline{w}^T \underline{x}^{(i)} + b) > 0$

$\xi^{(i)} = 0$ $\qquad$ $\xi^{(i)} = 1 - t^{(i)}(\underline{w}^T \underline{x}^{(i)} + b)$

$$\Rightarrow \xi^{(i)} = \max\left[0,\ 1 - t^{(i)}(\underline{w}^T \underline{x}^{(i)} + b)\right]$$

The penalty is the sum of $\xi^{(i)}$, so

$$L_H(\underline{w}, b) = \underbrace{\min_{\underline{w}, b} \sum_{i=1}^{N} \max(0, 1 - t^{(i)} y^{(i)}(\underline{w}, \underline{s}))}_{\text{Hinge-loss}} + \underbrace{\frac{1}{2\lambda} \|\underline{w}\|^2}_{\text{regularization}}$$

Soft margin SVM: linear classifier with Hinge loss and l_2 regularization

Dual:

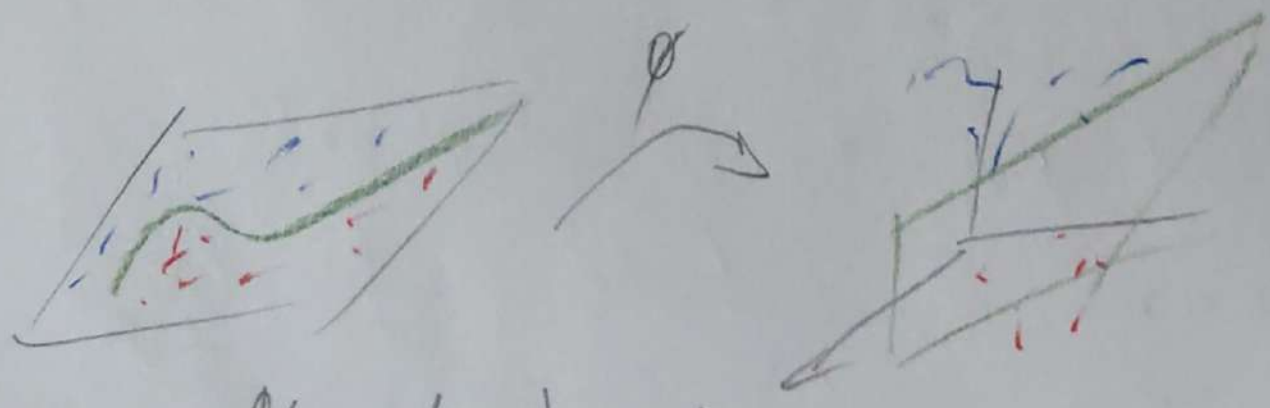
$$\text{maximize } f(c_1 \dots c_N) = \sum_{i=1}^{N} c_i - \frac{1}{2} \sum_{i,j=1}^{N} t_i t_j c_i c_j (\underline{x}_i^T \underline{x}_j)$$

$$\sum_{i=1}^{N} c_i t_i = 0 \qquad 0 \le c_i \le \frac{1}{2N\lambda} \quad \text{for } \forall i = 1 \dots N$$

$$\underline{w} = \sum_{i=1}^{N} c_i t_i \underline{x}_i$$

efficiently solvable by quadratic progr.

Kernel-trick for non-linear boundary



$\varphi(\underline{x})$ Kernel function

Trans: $\underline{x}_i \to \varphi(\underline{x}_i)$

$$f(c_1 \cdots c_N) = \sum_i c_i - \frac{1}{2} \sum_{i,j}^{N} t_i c_i \underbrace{\varphi(x_i)\varphi(x_j)}_{K(x_i, x_j)} t_j c_j$$

Example: $\varphi: \mathbb{R}^2 \to \mathbb{R}^6$

$$\varphi(x_1, x_2) = (1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, x_2^2, \sqrt{2}x_1 x_2)$$

$$\varphi^T(x_1^{(1)}, x_2^{(1)})\,\varphi(x_1^{(2)}, x_2^{(2)}) = (1 + x_1^{(1)} x_1^{(2)} + x_2^{(1)} x_2^{(2)})^2$$

$$K(\underline{x}^{(1)}, \underline{x}^{(2)}) = (1 + x_1^{(1)} x_1^{(2)} + x_2^{(1)} x_2^{(2)})^2$$

Summary

+ Works well with high dim data
+ Best if more features than data
+ Robust to overfitting
+ Kernel tricks for non-linear or non-separable data

- Not good for large datasets
- deep learning is better in finding kernels
- interpretability